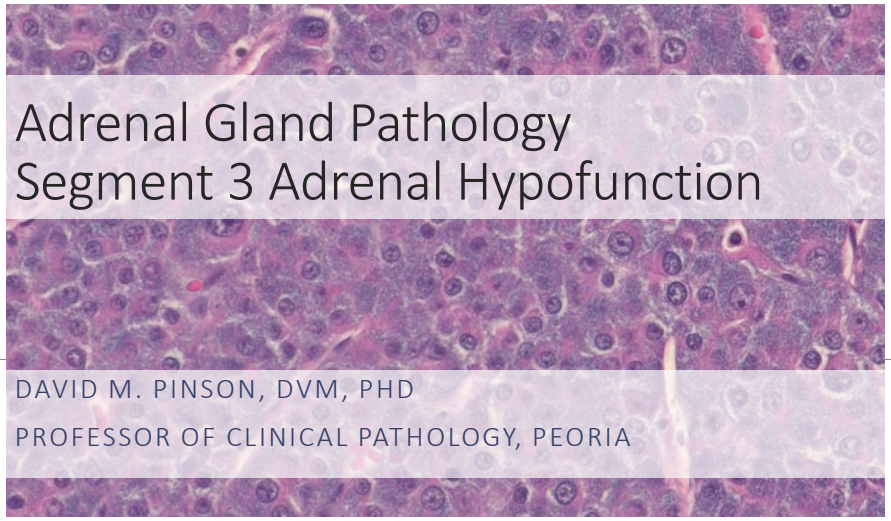


Adrenal Gland Pathology
Segment 3 Adrenal Hypofunction

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Greetings students and welcome to the third recording in this series.

Microlecture Series

Segment	Title	Duration
Segment 1	Overview	4 min 46 sec
Segment 2	Adrenal Hyperfunction	17 min 13 sec
Segment 3	**Adrenal Hypofunction**	**14 min 35 sec**
Segment 4	Adrenal Neoplasia	10 min 01 sec

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In this session, I'll discuss adrenal hypofunction, including deficiency of glucocorticoids, mineralocorticoids, and ACTH.

Learning Objectives

LO4 Evaluate the epidemiology, risk factors, and signs and symptoms of hypocortisolism.

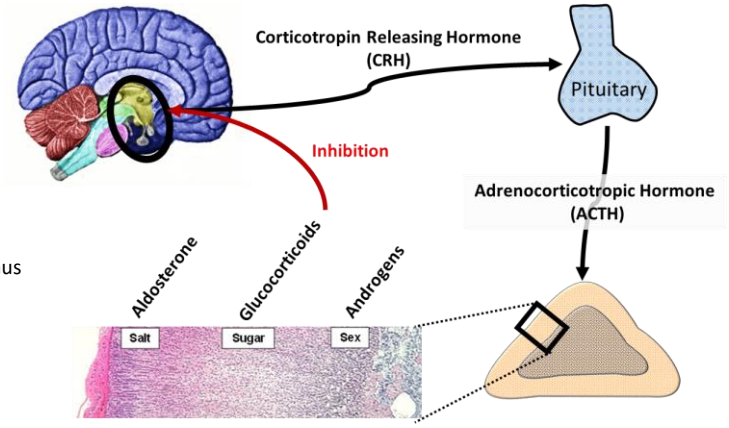
LO5 Describe the etiology, pathogenesis, and pathophysiology of Addison's disease, and Waterhouse-Friderichsen syndrome.

This session will cover the fourth and fifth learning objectives.

LO4 LO5

Adrenal Cortical Hypofunction

- Deficiency
  - Glucocorticoids
  - Mineralocorticoids
  - Both
- Site of Disease
  - Primary – adrenal gland
  - Secondary – Hypothalamus
  - Secondary – Pituitary
- Temporal Considerations
  - Acute syndrome
  - Chronic syndrome



Adrenal gland deficiency can involve the loss of glucocorticoids, mineralocorticoids of both. The loss of both cortisol and aldosterone is usually associated with destruction of the adrenal gland. This is referred to as primary hypofunction. Loss of cortisol can also be due to disease of the pituitary gland or the hypothalamus. This is termed secondary hypofunction. It is important to recognize that secondary hypofunction does not usually disrupt aldosterone physiology. This is an important feature distinguishing the two categories of adrenal hypofunction. Only primary adrenal gland destruction disrupts aldosterone. These syndromes can be acute with rapid onset or chronic and insidious onset.

LO4 LO5

Adrenal Cortical Hypofunction

- Acute syndromes
  - Abrupt withdrawal of exogenous steroid
  - Stress imposed on chronic adrenal insufficiency (adrenal crisis)
  - Massive adrenal destruction
    - Neonatal hypoxia
    - Post-operative DIC
    - Anticoagulant therapy
    - Waterhouse-Friderichsen syndrome

Chronic Steroid Consumption ↓↓ACTH



Cortical Atrophy



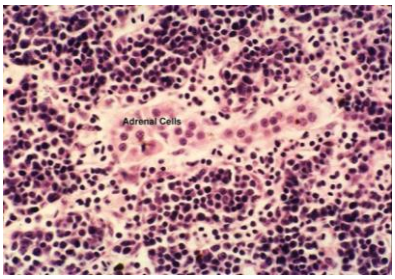
https://upload.wikimedia.org/wikipedia/commons/9/92/Sandoz.Methylprednisolone.4mg.jpg

Acute syndromes are associated with precipitous disease. They are typically rapid onset and a few examples are illustrated here. One of the more common problems is abrupt discontinuation of corticosteroid therapy. If these drugs are taken long enough, the adrenal gland becomes atrophic due to lack of ACTH. So, if exogenous corticosteroids are abruptly stopped, the patient can develop a precipitous syndrome of glucocorticoid deficiency. This is why most packaging of corticosteroids come in a tapering dose as shown in the picture. This permits a slow restoration of ACTH production and rebuilding of adrenal secretion under tropic stimulation. Another acute syndrome is a stress induced adrenal crisis in someone with marginal adrenal gland function or long term insidious disease. This usually occurs in patients with subclinical ongoing destruction of the adrenal gland. When 80 or 90% of the gland is gone, stressors can induce what is called an adrenal crisis. This is a very constitutional type of shock syndrome. The final category for acute syndromes involve acute massive destruction of the glands due to any of several causes. Waterhouse-Friderichsen is a good example and we'll discuss that syndrome in a few slides.

LO4 LO5

Adrenal Cortical Hypofunction

- Chronic Syndromes – 90% of cases due to these
  - Autoimmune adrenalitis; 60-80% of US cases
  - Autoimmune polyendocrine syndrome
    - Type 1
    - Type 2
  - Granulomatous inflammation
    - Tuberculosis - #1 worldwide cause
    - Sarcoidosis
  - AIDS
    - Opportunistic infection (CMV, MAC)
    - Kaposi's sarcoma
  - Metastatic cancer to adrenals
  - Genetic basis – rare



Autoimmune Adrenalitis

About 90% of the cases of adrenal gland deficiency are associated with chronic syndromes. These are fairly constitutional diseases. In fact, average patients go more than a year before a firm diagnosis is established. This can be quite challenging to recognize clinically. Autoimmune adrenalitis is the most prevalent cause in the US and it occurs in the context of some syndromes, particularly polyendocrine autoimmune syndromes 1 and 2. Other causes of adrenal gland destruction include tuberculosis, sarcoidosis, opportunistic infections, and metastatic cancer. The image shows the microscopic lesions of autoimmune adrenalitis. The gland architecture is replaced by lymphoid inflammation.

LO4 LO5

Autoimmune Polyendocrine Syndromes

- Syndromes of multiple endocrine dysfunction
- **Polyendocrine autoimmune syndrome 1**
  - Females more than males
  - Also known as autoimmune polyendocrinopathy-candidiasis-ectodermal dystrophy (APECED)
  - Mutations of the AIRE (autoimmune regulator) gene on 21q22.
  - Hypothyroidism, oral candidiasis, adrenal insufficiency (age 10-15)
  - Hypogonadism
  - Malabsorption and GI disorders
- Polyendocrine autoimmune syndrome 2

There are two syndromes that can be associated with autoimmune adrenalitis. These often include thyroid disease as part of the clinical expression. So polyendocrine autoimmune syndrome type 1 is due to a mutation in the autoimmune regulator gene, AIRE on chromosome 21. This syndrome is usually identified in children or adolescents and includes hypothyroidism, oral candida, adrenalitis, and a peculiar dermatopathy. GI disturbances can also be a part of this syndrome.

LO4 LO5

Autoimmune Polyendocrine Syndromes

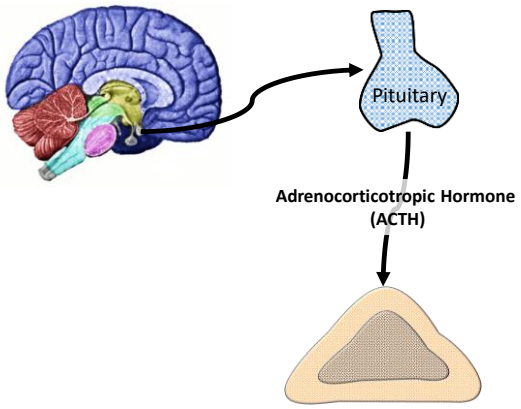
- Syndromes of multiple endocrine dysfunction
- Polyendocrine autoimmune syndrome 1
- **Polyendocrine autoimmune syndrome 2**
  - Much more common than syndrome 1
  - Half of the cases are familial
  - Adrenal insufficiency is the principal manifestation (50%)
  - Associated with autoimmune thyroid disease (Schmidt's Syndrome)
  - Associated with type 1 diabetes
  - Other non-endocrine problems: vitiligo, myasthenia gravis, Sjögren's, RA, etc.

Polyendocrine autoimmune syndrome 2 is much more common and includes adrenal insufficiency in approximately 50% of the cases. Other co-morbidities include Hashimoto thyroiditis and type 1 diabetes. It often occurs in the context of other autoimmune diseases. Adrenal gland destruction can occur in either syndrome. So Identification of thyroid disease should trigger some consideration of these syndromes and the potential for adrenal hypofunction or Addison disease.

LO4 LO5

Adrenal Cortical Hypofunction

- Secondary adrenal cortical insufficiency
- Reduced or no ACTH
- Hypothalamic or pituitary disease
  - Neoplasia
  - Infection
  - Sarcoidosis
  - Infarction
  - Irradiation



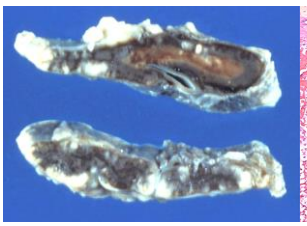
Adrenal gland hypofunction can be due to disease of the pituitary or hypothalamus. Essentially anything that destroys the pituitary can cause this. The disease is due to loss of ACTH. It is important to recognize that this loss of ACTH does not disrupt aldosterone physiology. Secondary hypocortisolism does not disrupt salt and hydrogen ion balance.

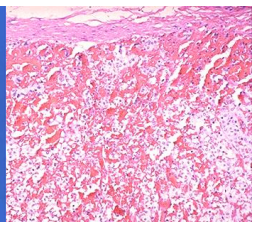
LO5

Waterhouse-Friderichsen Syndrome

- Classically associated with meningococcal infection in children and adolescents
- May be associated with Pseudomonas, Pneumococcus, Haemophilus, Staph
- Shock and DIC →
  - Hemorrhagic infarction of adrenals
  - Adrenal insufficiency
- Risk factors for adrenal hemorrhage
  - Thrombocytopenia
  - Heparin use
  - Sepsis







The Waterhouse-Friderichsen syndrome is one of the acute syndromes. It is classically associated with meningococcal infection in children and adolescents, but other organisms can cause the problem. The infection causes DIC and shock and results in hemorrhagic infarction of the adrenal gland. Since this is destruction of the gland, the syndrome is associated with loss of cortisol, a stress hormone, and aldosterone. Some of the risk factors that are associated with development of this complication of infection include thrombocytopenia, use heparin in DIC, and signs of systemic sepsis. The images show the more devastating side of this syndrome include skin necrosis and hemorrhagic infarction of the adrenal glands.

LO4 LO5

Chronic Adrenal Insufficiency - Clinical

- Disease at 90% destruction of glands
- Chronic Syndromes – Primary adrenal disease
  - Slow insidious onset
  - Weakness
  - Fatigue
  - Primary only, destruction of adrenal gland → Elevated ACTH→
    - Cutaneous hyperpigmentation
    - Oral hyperpigmentation
- Secondary → No ACTH elevation → no hyperpigmentation
  - No deficiency of aldosterone
  - No electrolyte abnormalities

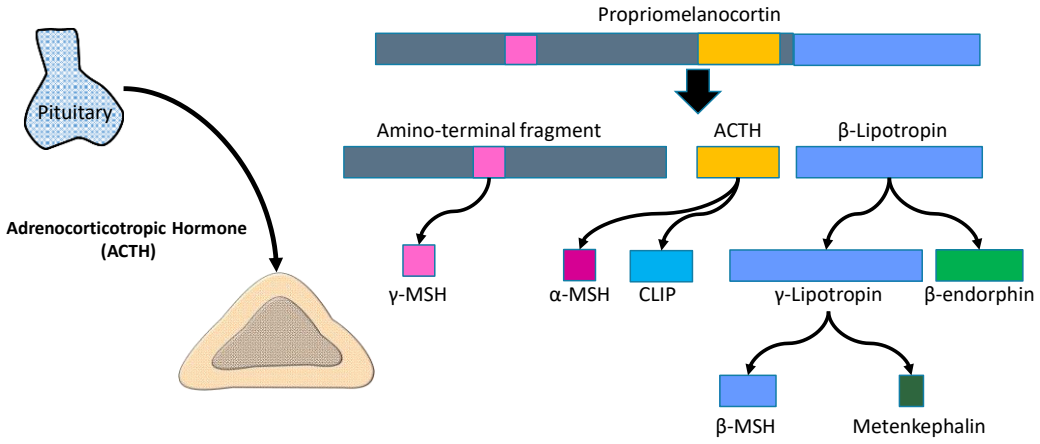




Chronic adrenal insufficiency or Addison disease is a more constitutional presentation. This includes weakness and fatigue, which come with a long list of differential diagnoses. A proportion of Addison disease patients show some pigmentation changes in the skin and oral cavity. This is only seen in primary adrenal gland disease since this causes elevated ACTH secretion. The pigmentation changes are only found when ACTH is elevated. Secondary hypocortisolism does not cause this problem. To complicate matters, not all patients with elevated ACTH will have the pigment changes.

LO4 LO5

Chronic Adrenal Insufficiency - Clinical

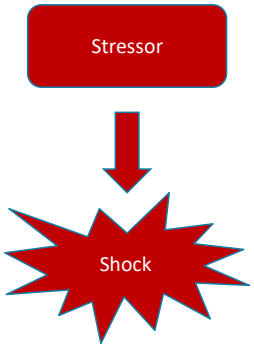


Why does elevated ACTH production cause pigmentation changes? It's because the production of the ACTH protein also results in production of melanocyte stimulating hormone or MSH. Processing of the propriomelanocortin RNA illustrates this co-production. So in some patients, increased production of ACTH is associated with increased production of MSH.

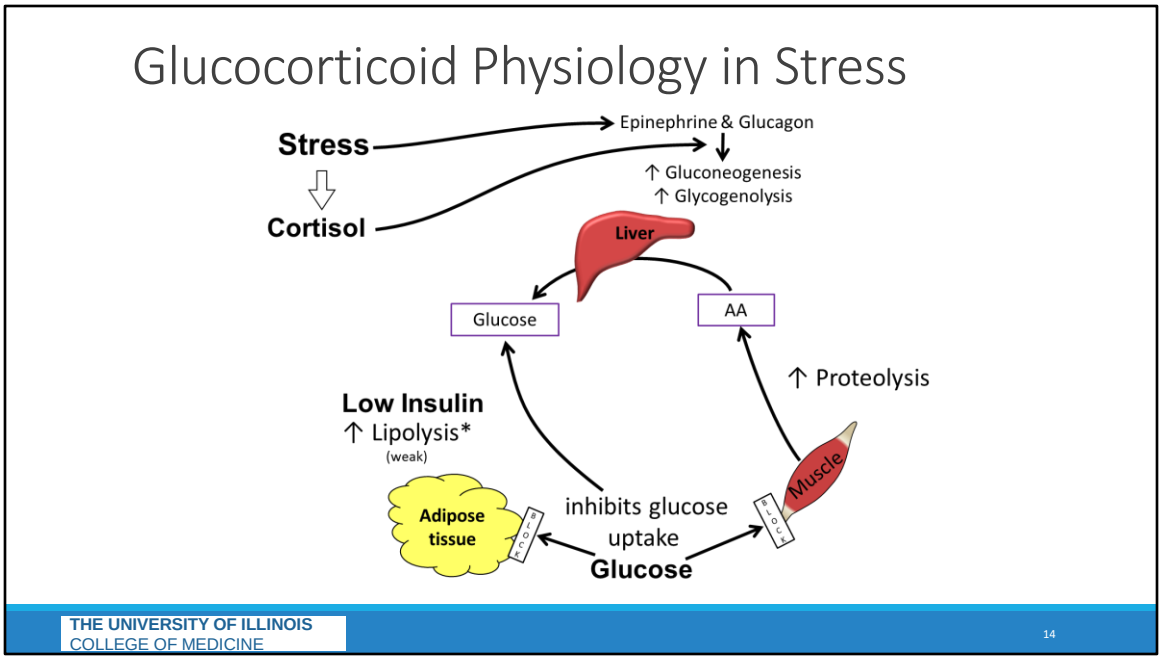
LO4 LO5

Chronic Adrenal Insufficiency - Clinical

- Acute Syndrome on chronic insufficiency – adrenal crisis
  - Often precipitated by a stressor
  - Shock
  - Weakness, fatigue
  - GI disturbances – nausea, vomiting, diarrhea
- Mineralocorticoid loss
  - Occurs ***only*** in primary disease; not secondary
  - Salt wasting
  - Renal tubular acidosis type IV
  - Electrolyte disturbances
    - Hyperkalemia
    - Hyponatremia

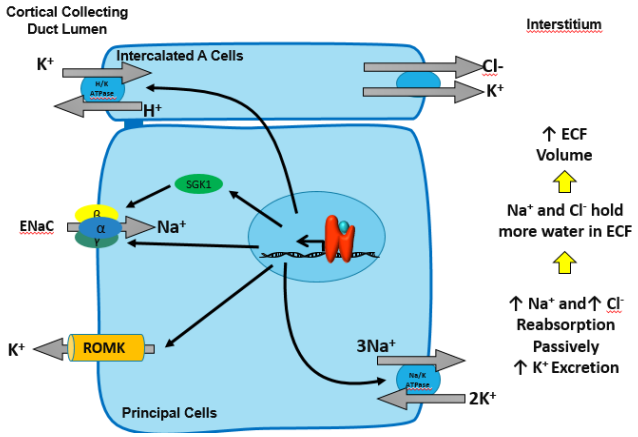


A unique aspect of slow insidious primary adrenal gland disease is the adrenal crisis or Addisonian crisis. This is often precipitated by a stressor, some type of trigger that stresses or challenges the patient. Because production of stress hormones is compromised, the patient cannot adapt well to the stressor. This induces precipitation of a shock syndrome with drop in blood pressure and often other nebulous signs such as nausea and vomiting and diarrhea. These clinical signs obviously can misdirect a clinician to numerous other differentials focused on the GI tract. There are however some unique features related to the loss of mineralocorticoids. Remember this only occurs in primary adrenal gland destruction syndromes, most commonly autoimmune adrenalitis. The loss of aldosterone causes alterations in sodium and potassium balance and hydrogen ion management. So your understanding of cortisol physiology, stress responses, and electrolyte balance will help you in understanding this disease.



Cortisol is a stress response hormone. It alters metabolism in favor of glucose production and reduces insulin responsiveness. So cortisol stimulates gluconeogenesis and glycogenolysis, both of which drive up glucose. This is a favorable response to stress.

Aldosterone Physiology

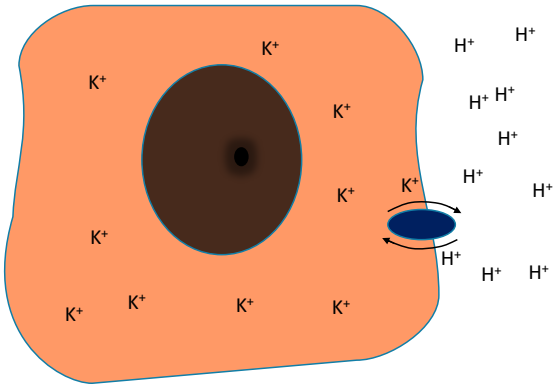


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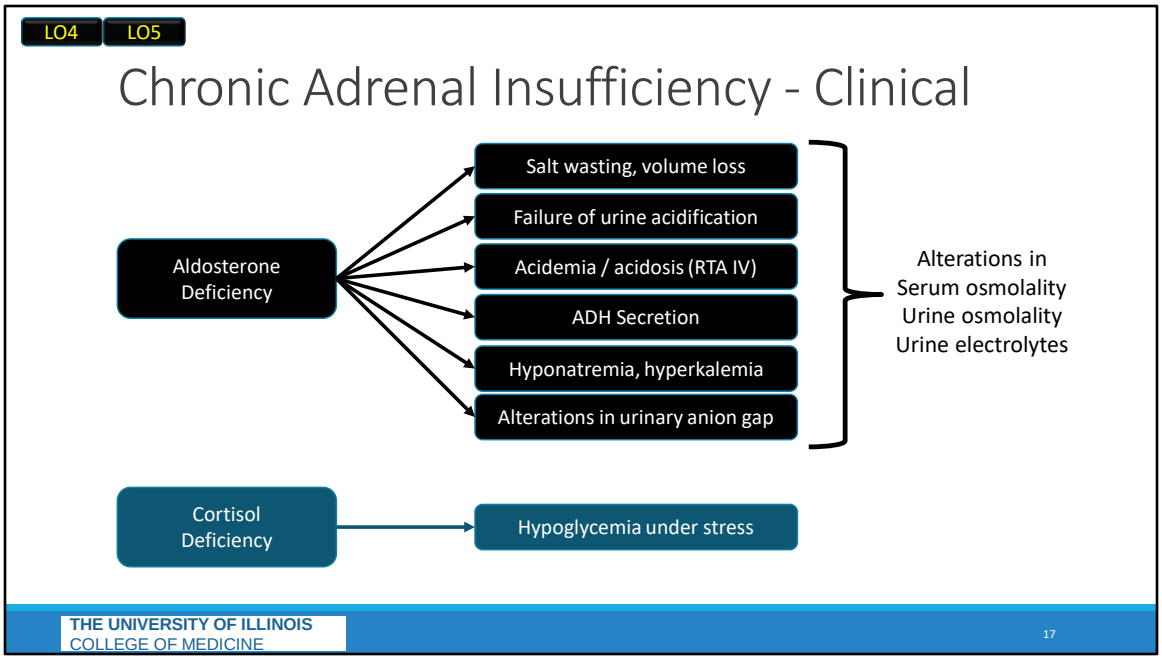
Aldosterone is lost with primary destruction of the adrenal cortex. Remember this does not happen in secondary adrenal hypofunction. Aldosterone functions in sodium reabsorption, potassium excretion, and hydrogen ion excretion. Now consider what happens if that hormone disappears. Sodium is lost. This is called salt wasting. Potassium is not excreted or essentially, it is retained. Labs will classically show hyponatremia and hyperkalemia. With salt wasting, water is lost. Hydrogen ions are not excreted. This causes a failure in urine acidification and retention of hydrogen ions. This causes acidemia and the syndrome is a type IV renal tubular acidosis. So disturbances in sodium, potassium and hydrogen ions should prompt a close look at aldosterone.

Ion Exchange in Acidemia



Increased Acid in blood
H+ exchanges for K+ in cells
Hyperkalemia

You should also think back about what happens in acidemia. This is a decrease in blood pH caused by an excess of H+ ions. In this condition, H+ ions are exchanged for K+ ions inside of cells. To some degree this mitigates the acidemia, but it also contributes to the hyperkalemia.

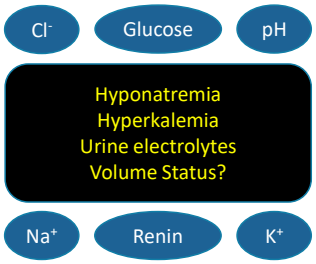


The syndrome of primary adrenal deficiency involves a loss of aldosterone and a loss of cortisol. The loss of aldosterone has a plethora of effects as noted on this slide. This results in alterations of serum electrolytes, fluid balance, urine electrolytes, and acid base balance. The electrolytes and fluid balance alterations can change plasma osmolality. The loss of effective circulating volume can result in ADH secretion. All of these analytes are measurable in the laboratory. Loss of cortisol may alter the patient's ability to maintain glucose under stress. This can cause hypoglycemia when it would not be appropriate.

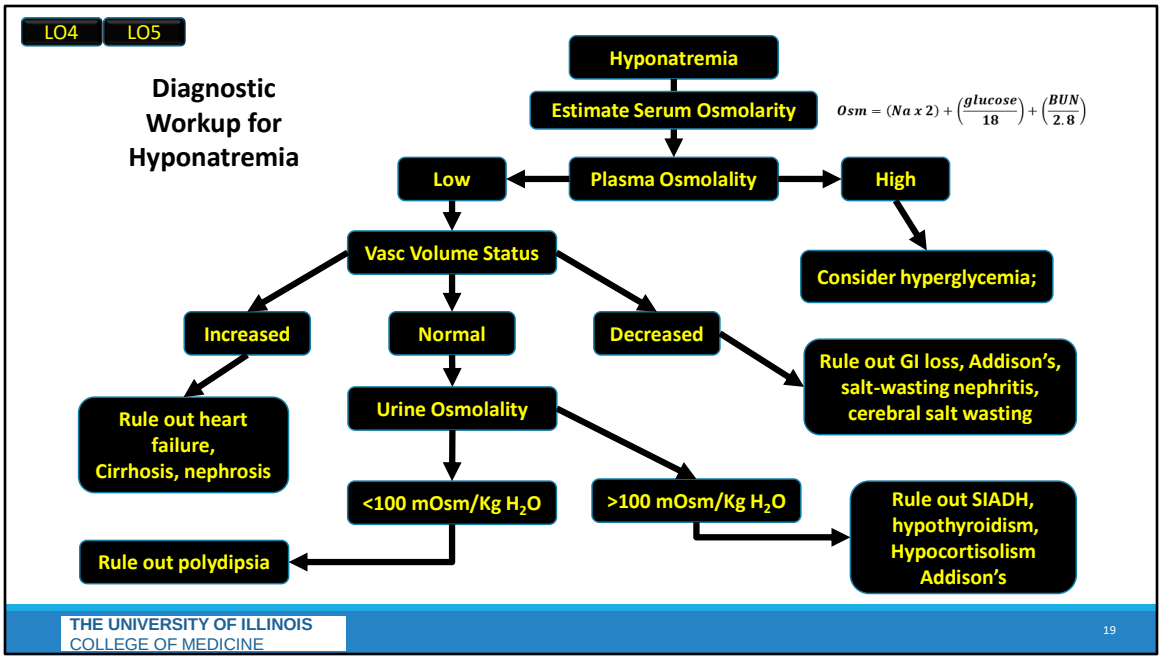
LO4 LO5

Laboratory Testing

- Serum glucose
- Serum aldosterone
- Serum renin
- Plasma osmolality
- Serum electrolytes
- Arterial blood gases
- Urine osmolality
- Urinary electrolytes
- **Labs must be interpreted with an understanding of patient volume status**



We can measure glucose, aldosterone, renin, serum and urine electrolytes, blood gasses, and serum and urine osmolality in the laboratory. The hallmarks of classical Addison disease due to adrenal gland destruction is hyponatremia and hyperkalemia, often with acidosis. There is a significant admonition. The interpretation of these findings is not simply based on reference ranges. The interpretation of renin, aldosterone, and ADH concentrations is heavily dependent on the condition of the patient. The data must be interpreted in the context of volume status. The serum concentration for renin, aldosterone and ADH can vary with volume status. This is complicated analysis that requires careful consideration of the patient and the data. An algorithm for this analysis is shown on the next slide.



Addison disease can present as hyponatremia. This algorithm is the clinical analysis of hyponatremia. There are other causes in addition to Addison disease. The identification of hyponatremia should immediately prompt an estimation of serum osmolarity. This is conveniently done with any of several published formulas or calculators. One such formula is shown. This is important for estimating whether it should be high or low based on the electrolytes that are responsible for most of the plasma solutes. The actual osmolality is then measured to determine how it compares to the estimated. If the measured osmolality is much higher, there is a hidden solute, often times glucose.

If the measured osmolality is low, then volume status has to be considered. Volume status can normal, high or low. If the volume status is high or increased, the differentials are those listed. Heart failure or cirrhosis are common causes.

If the volume status is decreased or low, then other differentials are considered. These include Addison disease, GI loss, and cerebral salt wasting.

If the volume status is normal, then evaluation of urine electrolytes becomes necessary. Here the urine osmolality is interpreted for differentials of excess fluid consumption versus salt wasting or other endocrine conditions such as Addison disease or syndrome of inappropriate ADH secretion. This illustrates why the interpretation of electrolytes, aldosterone, and renin are so complicated. The volume

status alters the interpretation of the analytes.
You also have to consider what the appropriate response of renin and aldosterone should be to conditions of ineffective circulating volume or fluid loss. The typical response is secretion of renin and eventually aldosterone. This should render the patient salt avid and the relevant finding is low salt in the urine. The problem in primary Addison disease is that the adrenal gland cannot produce the Aldosterone when it's needed.

Summary

- Primary adrenal insufficiency associated with disease of the adrenal gland.
- Secondary adrenal insufficiency due to pituitary or hypothalamic disease.
- Syndromes that can be acute (crisis, Waterhouse-Friderichsen) or chronic.
- Pigmentation alterations associated with elevated ACTH in primary disease
- The pathogenesis of adrenal crisis and its presentation as shock with electrolyte changes.
- Diagnostic algorithms for hyponatremia and the diagnosis of Addison disease.
- UP NEXT: A summary of adrenal tumors.

In this recording, I have summarized the loss of adrenal gland function as primary due to destruction of the gland or secondary due to pituitary or CNS disease. I have described these problems as acute and precipitous or chronic and insidious. I indicated how skin pigmentation may provide some clues, but not in all cases. I detailed the pathogenesis of an Addisonian crisis and explained the complexities of laboratory evaluation of electrolytes and acid base balance in the context of missing aldosterone. The last recording in this series will be on adrenal gland neoplasia.